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Incidence and pathophysiology of diabetes in South Asian adults living in India and Pakist an compared with US populations

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ABSTRACT & BACKGROUND

Objectives: To compare diabetes incidence, risk profiles, and pathophysiological drivers i n South Asian adults versus US populations.

Design: Multi-cohort prospective investigation evaluating incident diabetes across varied body mass index and demographic tiers.

Setting: Community surveillance across urban centers in Chennai and New Delhi, India, with parallel comparative US registries.

Participants: South Asian adults aged 20 years and older without baseline diabetes followe d across comprehensive clinical visits.

RESULTS: THE SOUTH ASIAN METABOLIC PHENOTYPE

Key Finding: South Asian adults exhibited higher diabetes incidence at markedly lower body mass index thresholds than Western counterparts.

Incident diabetes occurred in South Asians at a mean body mass index of 24.5 kg per square meter, within standard normal or overweight limits.

Beta-Cell Dynamics: Impaired early-phase insulin secretion was the predominant pathophysiological defect observed in South Asian participants.

Unlike Western populations where severe obesity-driven peripheral insulin resistance dominates, South Asians display early beta-cell exhaustion.

Visceral Adiposity: Despite lower overall weight, South Asian adults demonstrate elevated visceral fat and ectopic hepatic lipid accumulation.

This thin-fat metabolic profile necessitates diagnostic screening including body composition, visceral fat, and glycemic variation.

CLINICAL DISCUSSION & POPULATION SPECIFICITY

These findings underscore that universal Western body mass index cutoffs fail to capture m etabolic disease risk in Indian adults.

Metabolic intervention in Indian bodies must prioritize preserving lean skeletal muscle ma ss and reducing central visceral adiposity.

Dietary strategies must account for traditional Indian staple compositions, carbohydrate t hresholds, and sedentary occupational patterns.

Limitations: Observational design precludes definitive causal attribution; unmeasured diet ary and genetic factors may contribute.

Conclusion: Doctor-led metabolic health programs calibrated specifically for South Asian p hysiology offer superior preventive care.